PLL Random Jitter Estimation Using Different VCO Phase Noise Simulation Methodologies

Metha Jeeradit, Yohan Frans, Reza Navid, and Bruno Garlepp

Rambus Inc.

methaj@rambus.com, yfrans@rambus.com, rnavid@rambus.com, bgarlepp@rambus.com

ABSTRACT

Random jitter (RJ) is a significant noise component in PLL systems that use ring-based oscillators. In order to estimate RJ, accurate modeling of the VCO phase noise is essential. In this paper, the authors will present how the VCO phase noise they obtained from HSPICE RF and from the Impulse-Sensitivity Function (ISF) method compared to lab measurements, the limitations of the two methods and how these phase noise estimates can be used to obtain a prediction of RJ noise component in a PLL.

Table of Contents

1.0 Introduction 3
2.0 Estimating VCO Phase Noise 4
2.1 Brief Review of ISF Fundamentals 4
2.2 Estimating VCO Phase Noise with an ISF-based Methodology 4
2.2.1 ISF Simulation 4
2.2.2 Extracting Input Noise PSD 5
2.3 Estimating VCO Phase Noise with HSPICE RF 6
3.0 VCO Phase Noise Comparison 7
3.1 ISF-based and HSPICE RF Phase Noise Simulations vs Lab Measurement 7
3.2 Overlapping Regions between Low-Offset and High-Offset Frequencies in HSPICE RF 8
3.3 Limitations of ISF-based Methodology and HSPICE RF Phase Noise Simulator 10
4.0 Modeling Random Jitter in a PLL 11
4.1 Noise Sources in a PLL 11
4.2 From Input Noise to Output Jitter 12
4.3 Output Phase Noise and Jitter Comparison 13
5.0 Conclusions 15
6.0 Acknowledgements 15
7.0 References 15

Table of Figures

Figure 1-1: XAUI and CEI jitter specifications 3
Figure 2-1 Setup for ISF Simulation 5
Figure 2-2 Simulated ISF due to a single charge injection source and the corresponding oscillating waveform 5
Figure 2-3: Spice AC Noise Simulation Setup for a delay element 6
Figure 3-1: ISF-based and HSPICE RF Phase Noise Simulations vs Lab Measurement 7
Figure 3-2: Phase Noise Simulation Results from HSPICE RF for various Ring Oscillators 9
Figure 4-1: Noise Sources in a PLL 11
Figure 4-2: Measured and Approximated Reference Phase Noise Spectrum 11
Figure 4-3: Simplified Model for Estimating PLL Output RJ and Phase Noise 12
Figure 4-4: Simulated and Measured PLL Output Phase Noise for Several Damping Factor Settings 13
Figure 4-5: Simulated and Measured PLL Output RJ rms for Several Damping Factor Settings 14
Figure 4-6: Charge-Pump Noise Contribution to Output RJ 14

1.0 Introduction

Jitter in communication systems can be divided up into two components: Deterministic Jitter (DJ) and Random Jitter (RJ). Deterministic jitter can arise from many different sources such as supply sensitivity of various circuit elements, coupling from data signals, duty-cycle distortion, PLL charge-pump spurs, and ISI. Random jitter predominantly arises from the inherent noise of various PLL components such as the phase noise of the reference and main oscillators, and the thermal noise of the loop filter.

As can be seen in Figure 1-1 [1],[2], jitter specifications such as the XAUI and CEI standards place high importance on both DJ and RJ components. Hence, an accurate estimate of both components is essential to meeting the total jitter budget. This paper focuses on accurately estimating the RJ component in PLLs that use ring-based oscillators.

The main contributors to the output RJ of PLLs that use ring-based oscillators are reference noise and VCO noise. Reference noise contribution can only be minimized by adjusting the PLL's loop parameters. VCO noise can also be reduced by adjusting the PLL's loop parameters but also by modifying the design of the VCO circuit itself to reduce the phase noise it generates.

To estimate VCO noise contribution to a PLL's output RJ, an accurate phase noise simulation methodology is needed. This paper examines two different methods of estimating VCO phase noise and compare their accuracies in predicting PLL output RJ. The first method uses an Impulse-Sensitivity-Function (ISF) [3] to guide phase noise simulation while the second method uses the HSPICE RF simulator to simulate phase noise directly from standard device models and the VCO schematic netlist. This comparison is done for a 1.0-3.2GHz tunable PLL in a proprietary 90nm process.

Jitter Component	**XAUI**[1]	**CEI**
RJ (pk-pk)	0.18 UI @ BER = 1e-12	0.15 UI @ BER = 1e-12
DJ	0.17 UI[2]	0.15 UI

Figure 1-1: XAUI and CEI jitter specifications

[1] XAUI standard actually only specifies DJ (0.17UI) and TJ (0.35) requirements, hence 0.18 UI < RJ < 0.35 UI depending on DJ

[2] UI or Unit Interval is the period of a data bit

2.0 Estimating VCO Phase Noise

In this section, we first describe our methodology in estimating VCO phase noise using the ISF concept. Since our estimate will be compared with measured data, we include both the noise from the VCO delay elements and from the filter circuit that is attached to the VCO control voltage in our phase noise estimation. Then, we briefly explain how we estimate the same VCO phase noise components using HSPICE RF.

2.1 Brief Review of ISF Fundamentals

Let's first briefly review the concept behind Hajimiri's ISF [3]. The ISF-based methodology models oscillator phase noise as a linear but time-variant system. The input to the system (VCO) is a charge injected into an oscillating node and the output of the system is the resulting phase shift of the oscillating signal. The resulting input-output transfer function is cyclostationary and changes with the phase of the oscillating signal at the moment of injection but is linear at any given phase-instant. The Impulse-Sensitivity-Function (ISF) captures the behavior of the transfer function as a periodic time-varying, impulse response of the system and the total output phase shift is found by superpositionally integrating these time-varying impulse responses with the time-varying input.

Two sets of simulations are needed to estimate the phase noise: 1) ISF simulation and 2) input noise Power Spectral Density (PSD) simulation. Assuming that the resulting jitter is dominated by the phase noise in $1/f^2$ region – a usually valid assumption when the VCO is used in a PLL since the PLL filters out low frequency noise –the output phase noise is then given by [3]:

$$L(\Delta\omega) = 10\log_{10}\left(\frac{\Gamma_{rms}^2}{q_{\max}^2}\frac{\overline{i_n^2}/\Delta f}{2\Delta\omega^2}\right) \qquad \ldots (2.1)$$

Γ^2_{rms} represents the root-mean-square value of the ISF function. $i_n^2/\Delta f$ represents the input current noise PSD. q_{max} is the maximum charge-swing in a clock period. $\Delta\omega$ is the offset frequency. $L(\Delta\omega)$ is the output phase noise in dBc/Hz.

2.2 Estimating VCO Phase Noise with an ISF-based Methodology

2.2.1 ISF Simulation

ISF simulation is done by sweeping charge injection into an oscillating node across one clock period and measuring its output phase shift as shown in Figure 2-1 for our 4-stage VCO. Its simulated ISF due to a single source along with the corresponding oscillator waveform on the same node is shown in Figure 2-2. Typical characteristics of an ISF can be seen here where the oscillator is most sensitive during its crossing points while it is least sensitive during its peak and trough.

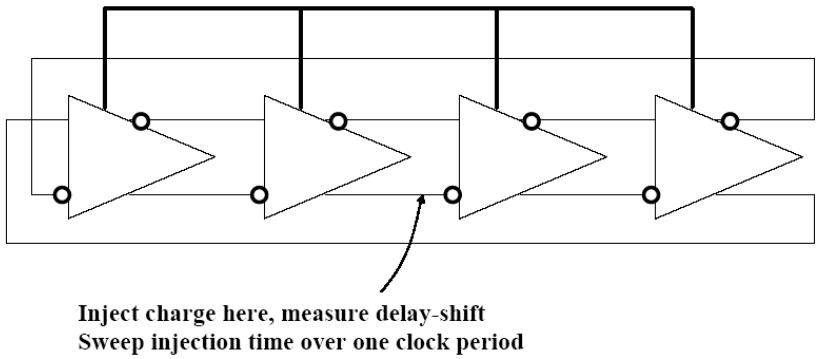


Figure 2-1 Setup for ISF Simulation

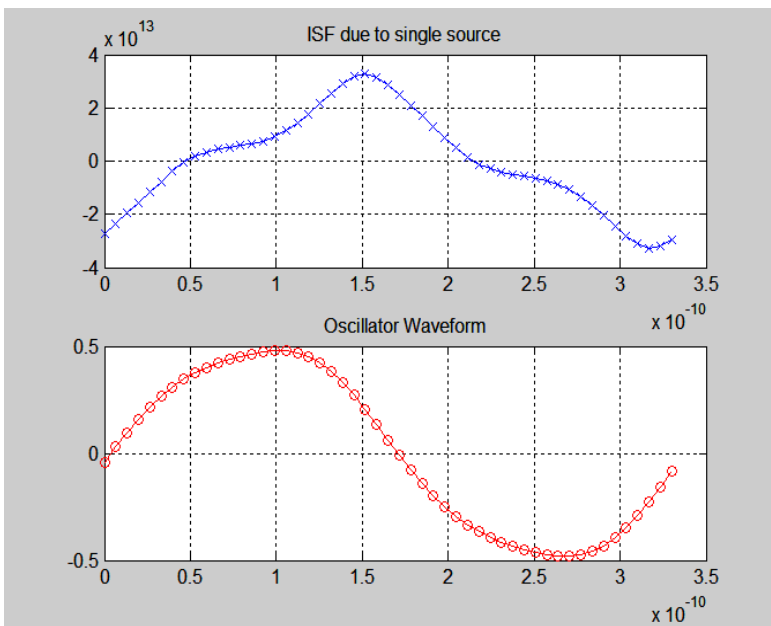


Figure 2-2 Simulated ISF due to a single charge injection source and the corresponding oscillating waveform

To obtain the overall ISF, we then assume that the noise sources injected on to different oscillating nodes are uncorrelated and that each node has the same rms ISF response due to the symmetry of the ring oscillator. We can then sum their ISFs in a root-mean-square fashion:

$$\Gamma_{total} = \sqrt{N}\Gamma_{indiv} \qquad \ldots (2.2)$$

To ensure that the simulated ISF is valid, we also check its linearity by varying the magnitude of the injected charge and making sure that the resulting output shift changes proportionally with the input magnitude [4].

2.2.2 Extracting Input Noise PSD

Next, we find the PSD of the input noise due to the delay elements. We bias a single delay element at its switching point as shown in Figure 2-3 and perform a spice AC noise simulation from DC to frequency offset of interest to obtain output referred current noise. The noise sources from each delay element are again assumed to be uncorrelated and are summed in a root-mean-square fashion.

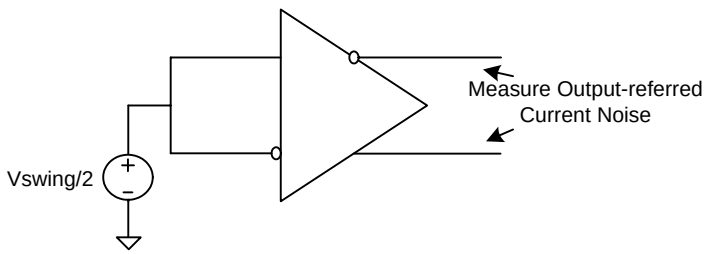


Figure 2-3: Spice AC Noise Simulation Setup for a delay element

As mentioned earlier, we also need to take into account the noise from the filter circuit that is connected directly to the VCO control voltage. In this paper, we include the VCO output noise from this source as VCO phase noise because this noise also contributes to the open-loop VCO noise that we measure in the lab. To simulate filter noise, we again bias our filter circuit at its operating condition and perform a spice AC noise simulation to obtain output referred noise.

We use this ISF-based methodology for estimating phase noise because of the simplicity of the simulations involved. To keep the simulations simple, we have assumed that the noise from the delay element is time-invariant and is represented by the worst-case noise, which occurs at the switching points of the VCO delay elements. Hence, we expect our ISF-based estimate to be slightly more conservative than our HSPICE RF estimate and the actual measurement results.

2.3 Estimating VCO Phase Noise with HSPICE RF

Estimating VCO phase noise with HSPICE RF is relatively simpler than using the ISF-based methodology described above. We set up both the filter circuit and VCO at their operating conditions and use the phase noise simulation feature of HSPICE RF [5]. Specifically, we used .HBOSC command to help find the oscillating frequency first then .PHASENOISE command to simulate the phase noise as shown below:

```
.HBOSC tones=fosc nharms=50
+PROBENODE = xvco.q0, xvco.q0b, 'Amplitude'
.PHASENOISE v(xvco.q0,xvco.q0b) dec 10 10k 10e8
```

f_{osc} is the oscillating frequency. `nharms` is the number of harmonics to simulate. `xvco.q0, xvco.q0b` are the differential oscillating node in our circuit. `Amplitude` is the expected oscillating amplitude of the VCO.

3.0 VCO Phase Noise Comparison

3.1 ISF-based and HSPICE RF Phase Noise Simulations vs Lab Measurement

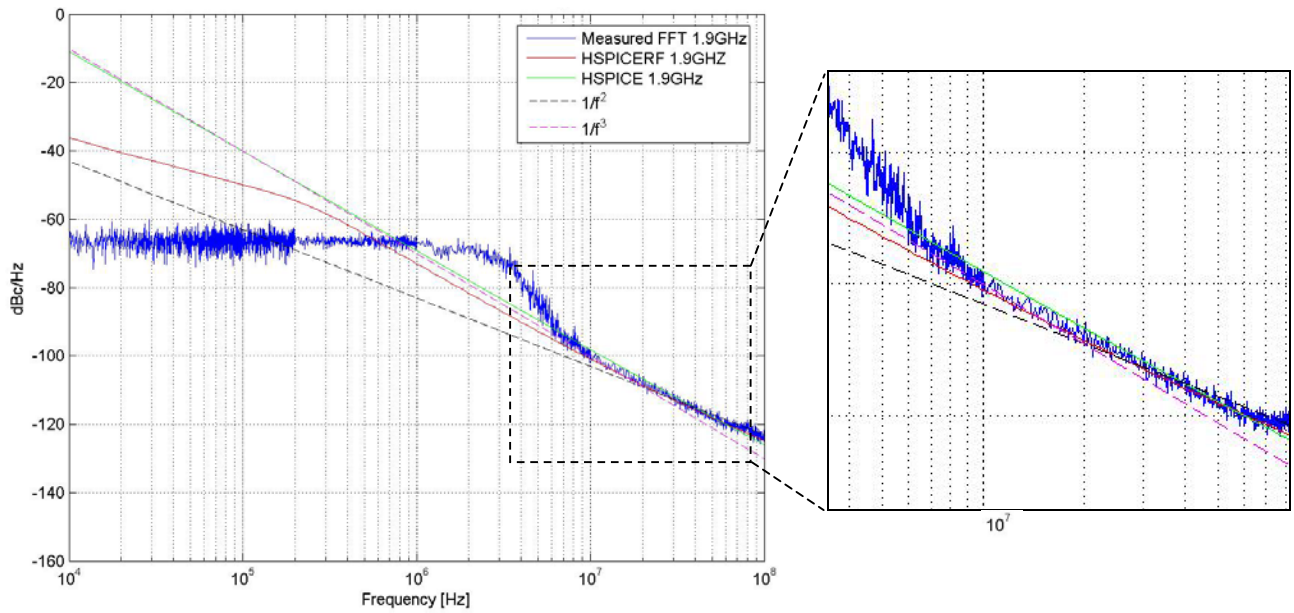


Figure 3-1: ISF-based and HSPICE RF Phase Noise Simulations vs Lab Measurement

Figure 3-1 shows the simulation results from ISF-based methodology and from the phase noise feature of HSPICE RF against lab measurement at 1.9GHz. From this graph, we can observe the following:

- Both simulation methods are reasonably accurate at high offset frequencies, e.g., above 10MHz. Since our PLL's loop bandwidth is around 20 MHz, this accuracy is good enough to predict overall phase noise of the PLL as will be shown later in Section 4.
- We do not know the cause of the steep-slope-region around 3-7 MHz, but the plateau at lower frequencies is likely due to resolution bandwidth of the spectrum analyzer.
- ISF-based simulation (labeled HSPICE 1.9GHz) slightly overestimates the phase noise compared to HSPICE RF simulation. This is likely due to our assumption of the time-invariant input noise PSD represented by the worst-case noise at the switching point.
- HSPICE RF and lab measurement show similar $1/f^3$ corners at around 20 MHz.

3.2 Overlapping Regions between Low-Offset and High-Offset Frequencies in HSPICE RF

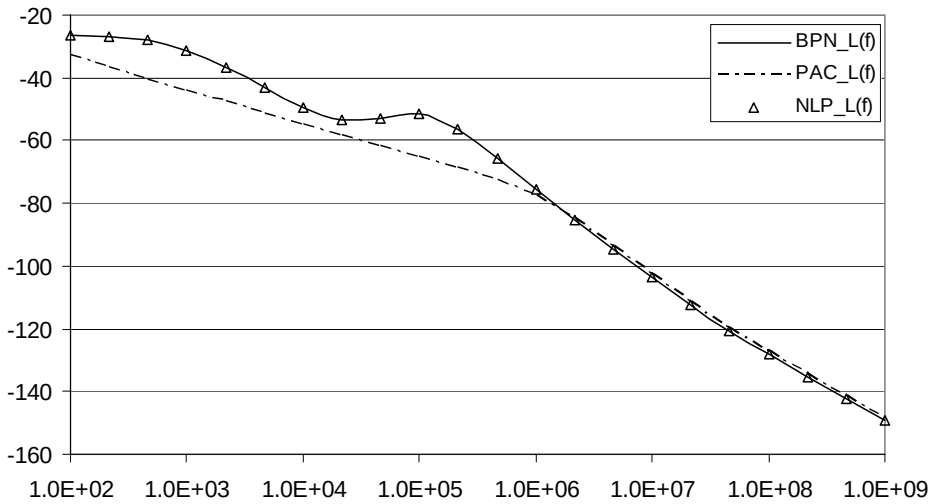
Although a high-level simulator such as HSPICE RF greatly facilitates phase noise estimation, its results should be used with care. According to HSPICE RF manual [5], the simulator is capable of calculating phase noise using two different algorithms: nonlinear perturbation (NLP) and periodic AC (PAC). Because of the underlying assumptions in each of these methods, neither of these methods is appropriate for calculating phase noise in a wide offset frequency band. Namely, NLP is said to be appropriate for small offset frequencies while PAC is suitable for large offset frequencies. Unfortunately, the definitions of large and small offset frequencies can be unclear as they depend on oscillator topology and device models. This causes uncertainty as to which method is appropriate for a given offset frequency. To circumvent this problem, HSPICE RF provides a third option in which the simulator calculates phase noise using both NLP and PAC. Subsequently, the simulator finds a middle region where both of these methods provide the accurate result (the two curves overlap). The simulator then provides a phase noise graph whose value is equal to the number given by the NLP for the offset frequencies smaller than the overlap region and by PAC for the offset frequencies larger than that region. This scheme is intended to provide a wide-band phase noise graph which is valid for the entire region of interest and is referred to as BPN in HSPICE RF.

In some situation, however, the two solutions provided by NLP and PAC do not have an overlap region. In such situations, HSPICE RF chooses the solution given by the NLP as the BPN solution for the entire band of interest. In some cases, this can lead to erroneous results for large offset frequencies. Incidentally, this offset frequency range is of most interest in closed-loop systems.

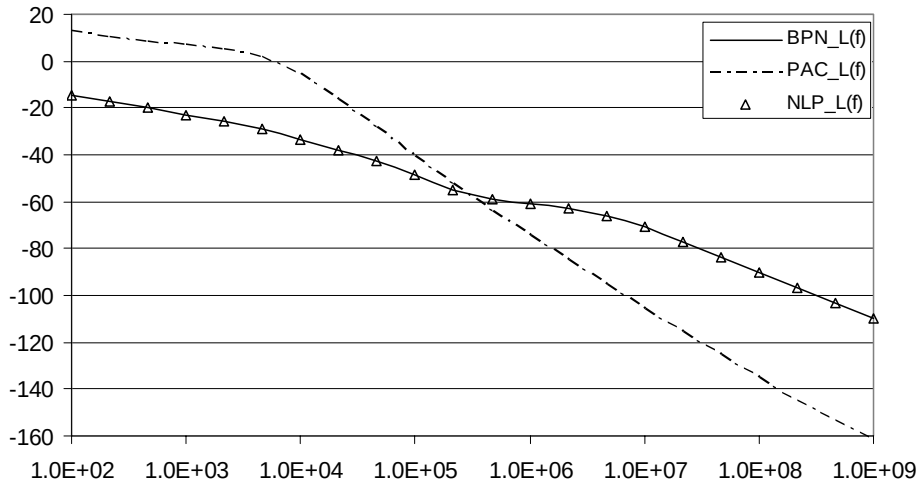
This is depicted graphically in Figure 3-2. This figure shows the phase noise graphs generated by HSPISCE RF for three different ring oscillators. In all case the NLP result is drawn with triangular markers, the PAC result in dash-dot lines and the final BPN phase noise graph in solid lines. Figure 3-2a presents the phase noise simulation result for a simple five-stage single-ended ring oscillator oscillating at 9.3GHz built in a 90nm bulk CMOS process. As can be seen in this case, the solutions given by the NLP and PAC methods do not have an overlap region and therefore the final phase noise graph is the same as the NLP solution. Luckily in this case, the difference between the NLP and PAC result at large offset frequencies is in the order of 1dB and therefore the error is relatively small.

This is not however always the case. Figure 3-2b shows phase noise simulation results for a four-stage pseudo differential ring oscillator oscillating at 2.5GHz built using floating body devices in a 65nm SOI process. As can be seen in this case, the solutions given by the NLP and PAC do not show an overlap region and therefore the NLP result is presented as the final phase noise graph. Unfortunately in this case, the difference between NLP and PAC results is very large at large offset frequencies resulting in unacceptable error.

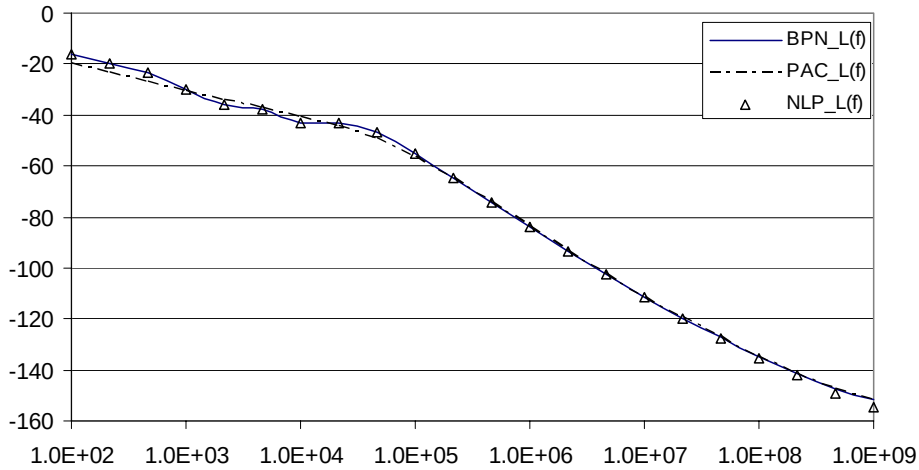
In order to find out the cause of this problem, we rebuilt this oscillator in a 90nm bulk CMOS process (the oscillation frequency reduces to 1.4GHz). Figure 3-2c shows phase noise simulation results for this oscillator. As can be seen, the two solutions of NLP and PAC do show an overlap



(a)



(b)



(c)

Figure 3-2: Phase Noise Simulation Results from HSPICE RF for various Ring Oscillators

region in this case and the final graph coincides with the NLP solution at small offset frequencies and PAC for large offset frequencies. Based on the above we suspect that the large error associated with the simulation result presented in Fig. 3-2b is related to the SOI process models.

To further understand the issue, we simulated the SOI oscillator after increasing the number of harmonics from 64 to 256 and setting the parameter PHNOISE_LORENTZ=0 (these changes were based on recommendations from HSPICE RF experts in Synopsys). This last simulation showed an overlap region for PAC and NLP algorithms at very low frequencies confirming that the major cause of error in this case are most likely the low-frequency poles associated with the floating body nodes in the SOI process.

Our conclusion is that although use of a high-level simulator such as HSPICE-RF greatly facilitates phase noise estimation, careful inspection of the final results and close interaction with simulator experts are recommended in order to avoid errors and to obtain the best results.

3.3 Limitations of ISF-based Methodology and HSPICE RF Phase Noise Simulator

Although both the ISF-based methodology and the HSPICE RF phase noise simulator provide comparable accuracy in estimating VCO phase noise, there are a few advantages and disadvantages to using each method. The advantages in using HSPICE RF are:

- HSPICE RF is more accurate and simpler to use:
    - We made some assumptions in ISF-based methodology (see Section 2.2) to keep the set of needed simulations needed small thereby sacrificing some accuracy, and
    - We applied equation 2.1 directly to the simulated noise input spectrum in the ISF-based methodology to estimate VCO output phase noise though the equation is actually only valid for white noise[3].
- Sensitivity analysis such as phase noise variation to control voltage is easily done in HSPICE RF by a SWEEP command.
- HSPICE RF provides accurate estimate of $1/f^3$ corner frequency. Although this information may not be directly useful to predicting jitter, it can help provide insights into actual VCO design.

On the other hand, the advantages in using the ISF-based methodology are:

- This methodology forces one to simulate filter-generated and actual VCO-generated phase noise separately which can provide helpful design insights into the dominating noise source.
- The ISF-based Methodology avoids the need for paying careful attention to selecting between NLP and PAC Simulation methods in HSPICE RF for overlapping regions between low-offset and high-offset frequencies as discussed in Section 3.2.

[3] Note that this is not a limitation of the ISF concept itself – a valid equation for 1/f noise can be found in [3] – but its usage in our simulation environment.

4.0 Modeling Random Jitter in a PLL

4.1 Noise Sources in a PLL

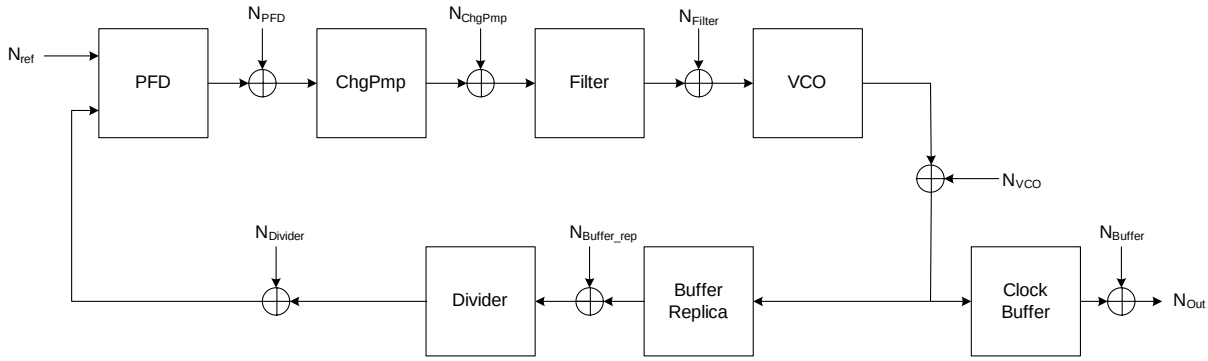


Figure 4-1: Noise Sources in a PLL

Figure 4-1 shows the noise sources in our PLL architecture. The three most significant noise sources are the reference noise (N_{ref}), filter (N_{Filter}) and VCO noise (N_{VCO}). The reference noise used in our model is two-segment, straight line function derived from data taken from lab measurements of a reference clock source of interest as shown in Figure 4-2. VCO and Filter Noise are estimated either from the ISF-based methodology or from HSPICE RF Simulations.

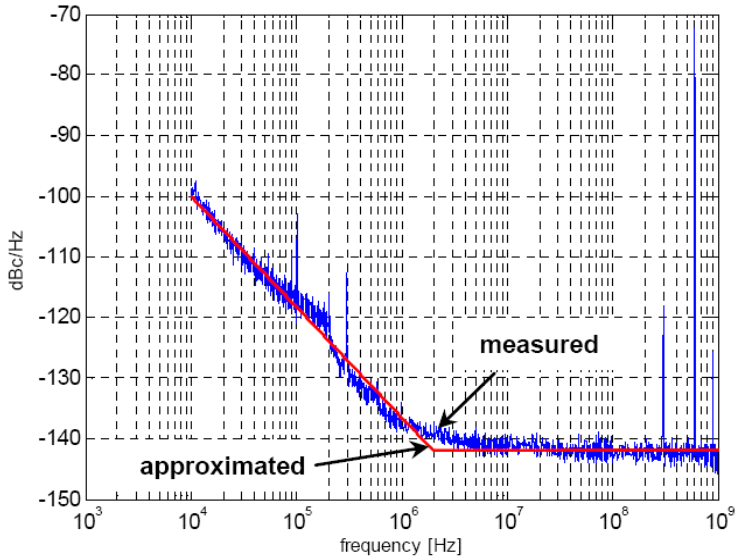


Figure 4-2: Measured and Approximated Reference Phase Noise Spectrum

The other noise sources – N_{PFD}, N_{ChgPmp}, $N_{Divider}$, N_{Buffer_rep} and N_{Buffer} – were found to be relatively insignificant primarily because they are gated noise and are only active for a short period of time during each cycle. These noise sources were estimated by biasing their associated circuits at their operating point and performing a spice AC noise simulation to obtain a raw output noise estimate. Then, the raw output estimate is convolved with a pulse train to account

for the sampled nature of the system to obtain an estimate of the effective input noise. Finally, this input noise is passed through the PLL transfer function to the PLL output (as described in the next section) to be compared against the output noise generated by the other sources.

4.2 From Input Noise to Output Jitter

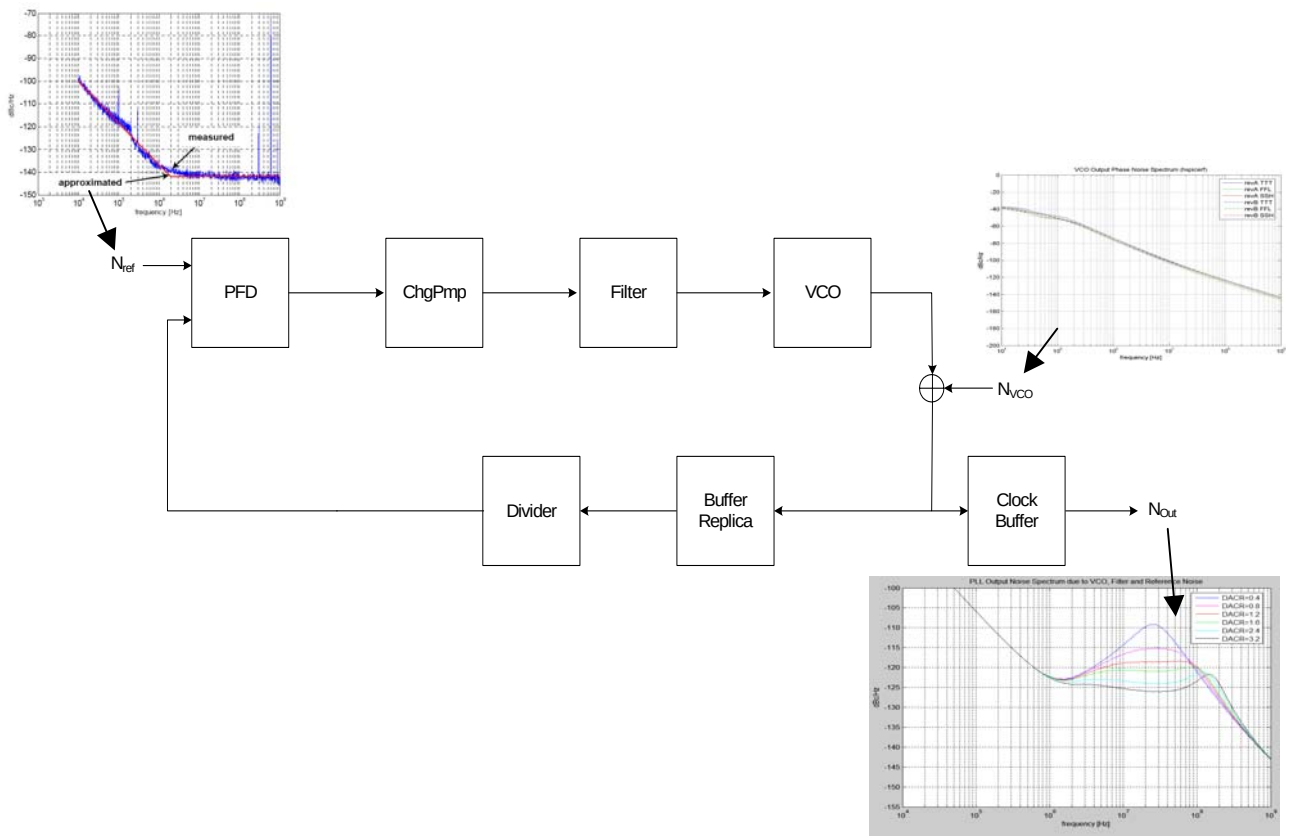


Figure 4-3: Simplified Model for Estimating PLL Output RJ and Phase Noise

Figure 4-3 shows the simplified model for estimating PLL Output RJ and Phase Noise. Only the three significant noise sources remain: reference, filter and VCO noise. As mentioned earlier, filter and VCO noise are estimated either by the ISF-based methodology or by HSPICE RF simulation and are referred as one single input to the PLL at the output of the VCO. To obtain an estimate of PLL output phase noise, the two inputs N_{ref} and N_{VCO} are passed through the square of their noise-to-output transfer functions and summed in uncorrelated fashion:

$$\begin{aligned} N_{Out} &= N_{ref}\left|H_{ref->out}(s)\right|^2 + N_{VCO}\left|H_{VCO->out}(s)\right|^2 \\ &= N_{ref->Out} + N_{VCO->Out} \end{aligned} \qquad \text{... (4.3)}$$

The output jitter is then given by the Wiener-Khinchine Theorem:

$$\sigma^2_{jitter} = \int_f N_{Out}(f)df \qquad \ldots(4.4)$$

Note that we could also find the jitter due to each noise source first by employing The Wiener-Khinchine theorem on $N_{ref\text{->}Out}$ and $N_{VCO\text{->}Out}$ separately and then sum their contributions in a root-mean-square fashion. The advantage of this approach is the available breakdown information on the relative contribution of the two sources.

4.3 Output Phase Noise and Jitter Comparison

Figure 4-4 shows the comparison of the output phase noise between the simulations and the lab measurement for several damping factor settings. We have two charge-pumps in our PLLs: an integrating charge-pump (DACC) and a proportional charge-pump (DACR). Both charge-pumps are adjustable and the damping factor is changed by varying the proportional charge-pump DACR. We can see from Figure 4-4 that both simulation methods provide good accuracy to the lab measurement over various damping factor settings with ISF-based methodology (HSPICE) slightly overestimating the total phase noise than HSPICE RF-based.

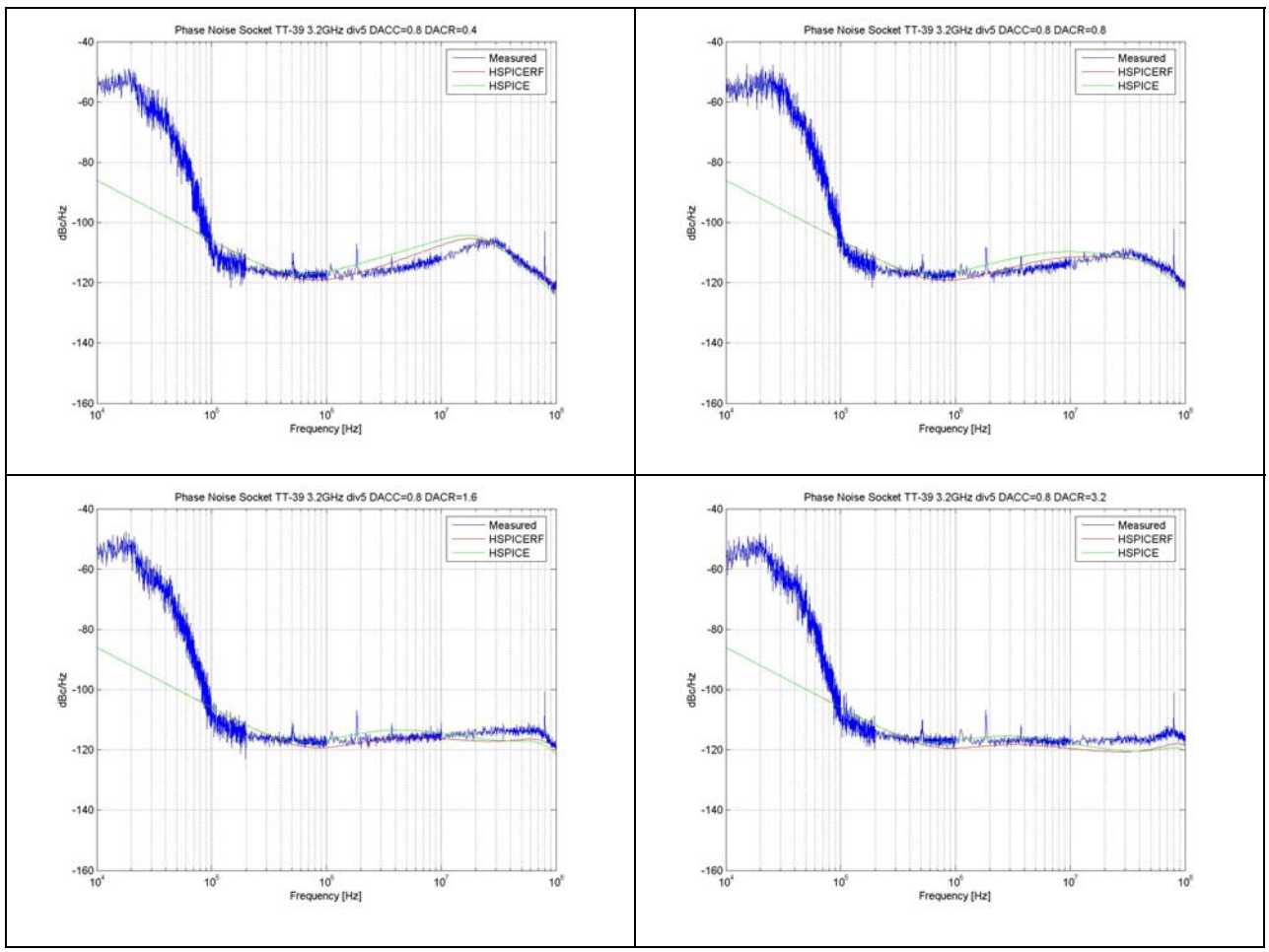


Figure 4-4: Simulated and Measured PLL Output Phase Noise for Several Damping Factor Settings

DACC/DACR	RJ Source	HSpice [mUI rms]	HSPICE RF [mUI rms]	Lab
0.8/0.4	Reference	2.5	2.5	
	Filter + VCO	14.3	13.1	
	Total	***14.5***	***13.4***	***11.8***
0.8/0.8	Reference	2.5	2.5	
	Filter + VCO	9.7	9.3	
	Total	***10.0***	***10.0***	***9.5***
0.8/1.6	Reference	2.6	2.6	
	Filter + VCO	6.8	6.8	
	Total	***7.3***	***7.3***	***7.9***
0.8/3.2	Reference	2.8	2.8	
	Filter + VCO	5.4	5.2	
	Total	***6.1***	***5.9***	***6.4***

Figure 4-5: Simulated and Measured PLL Output RJ rms for Several Damping Factor Settings

Figure 4-5 shows the corresponding comparison in jitter domain. As can be seen in all DACC/DACR combinations, the Filter + VCO contribution to RJ is always more significant than the contribution from the reference clock emphasizing the importance in having an accurate methodology to estimate VCO phase noise.

Apart from the DACC=0.8, DACR=0.4 settings, the accuracy of both ISF-based Methodology (HSPICE) and HSPICE RF-based are accurate within 10% of the lab measurements. The DACC=0.8, DACR=0.4 settings correspond to the top left graph in Figure 4-4 where the PLL has lowest damping factor or the most peaky response. We can see from this graph that the large jitter error in this case is due to the estimation error in the location of the peak frequency. To improve the accuracy in this scenario, a calibration of the PLL loop transfer function is first required. However, even without this calibration, the two estimation methodologies provide good accuracy in predicting PLL RJ output.

To put the contribution of the insignificant noise sources into context, Figure 4-6 shows the estimated RJ due to the two charge-pumps. The second column shows the cycle time of the reference clock while the third column shows the length of time that the charge-pump is active and the fourth column shows the estimated output RJ in mUI rms. Each charge-pump typically has about 100ps active period which contributes only 0.26 to 0.31mUI rms to the RJ output of the PLL. In the extreme case where the charge-pumps were fully on all the time, their contributions can become significant at 3.9 and 4.7mUI rms.

Source	Tref	Tactive	Estimated RJ rms (mUI)
Integrating Charge-Pump	1.56ns	100ps	0.26
	1.56ns	1.56ns	3.9
Proportional Charge-Pump	1.56ns	100ps	0.31
	1.56ns	1.56ns	4.7

Figure 4-6: Charge-Pump Noise Contribution to Output RJ

5.0 Conclusions

This paper compared two methods of estimating VCO phase noise and their accuracies in estimating PLL RJ output. An accurate VCO phase noise modeling is essential since this noise component typically dominates the total PLL output phase noise for architectures that use ring-based oscillators.

We have shown that both ISF-based methodology and HSPICE RF simulation provide good enough accuracy to predict PLL RJ. However, due to ease-of-use, an RF-based simulator such as the HSPICE RF is recommended over HSPICE-based method such as the ISF methodology described here as long as careful attention is paid to ensure that the resulting simulation is valid.

6.0 Acknowledgements

The authors would like to thank Carl Werner, Wendem Beyene, Barry Daly, Glen Chiu, Lei Luo, Trey Greer and Jared Zerbe at Rambus for their discussions and insights on estimating VCO phase noise and modeling RJ in a PLL. The authors would also like to thank Randy Nubling, Harald Von Sosen, Scott Wedge, and Sumit Vishwakarma at Synopsys for their excellent support on HSPICE RF tool as well as their insights and feedbacks on this paper.

7.0 References

[1] IEEE Standard 802.3ae clause 47.
[2] "Common Electrical Interface, revision OIF-CEI-02.0", Optical Internetworking Forum, Feb. 2005.
[3] A. Hajimiri. "Jitter and Phase Noise in Electrical Oscillators," Ph. D. Thesis, Stanford University, Nov. 1998.
[4] T. H. Lee. "Planar Microwave Engineering," Cambridge University Press, Oct. 04
[5] HSPICE RF User Guide, Version Y-2006.03, Synopsys, Mar. 06